

Kazakh British Technical University

Information communication technology

Laboratory Work №1

ERD diagram

Full Name: Ospan Ulzhan Muratkyzy

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Almaty

The designed ER Diagram represents a simplified International Airport Database system. It manages core airport operations, including flight scheduling, airline information, passenger profiles, bookings, and baggage handling. The database ensures data integrity by enforcing primary and foreign key constraints across multiple interconnected entities.

One-to-Many (1:M) Relationships:

Airline to Flight: One airline company can operate many flights, but each flight belongs to one specific airline.

Airport to Flight: One airport can serve as the departure for many flights, and as the arrival for many flights. Each flight has exactly one departure and one arrival airport.

Flight to Booking: A single flight can have multiple passenger bookings, but each booking is tied to one specific flight.

Passenger to Booking: A single passenger can make multiple bookings, but each booking belongs to one passenger.

Booking to Booking_Change: One booking can have zero or multiple changes logged in the system.

Booking to Baggage: One booking can have multiple pieces of registered baggage.

Booking to Baggage_Check: Baggage associated with a booking can undergo multiple checks.

Passenger to Baggage_Check: A passenger is the owner of the baggage being checked.

Passenger to Security_Check: A passenger can undergo multiple security checks over time.

One-to-One (1:1) Relationships:

Booking to Boarding_Pass: Each booking (for a specific seat) issues exactly one boarding pass, and each boarding pass corresponds to exactly one booking.

The provided ER Diagram uses standard relational database notations (Crow's Foot notation):

PK (Primary Key): Uniquely identifies each record in a table.

FK (Foreign Key): A field that links to the primary key of another table to establish a relationship.

|| (Two vertical lines): Indicates a "One and only one" relationship.

|< (Crow's foot with a line): Indicates a "One-to-Many" relationship.

